Series Convergence & Divergence Cheat Sheet

Calculus II Quick Reference Guide | Contains tests, power series, and error bounds

Basics & Definitions

Sequence: An ordered list of numbers $\{a_n\}$.
Series: Sum of a sequence $\sum_{n=1}^{\infty} a_n$.
Partial Sum (S_k): $S_k = \sum_{n=1}^{k} a_n$.

A series **converges** to S if $\lim_{k\to\infty} S_k = S$ (finite). Otherwise, it **diverges**.

The n-th Term Test (Divergence Test)

If $\lim_{n\to\infty} a_n \neq 0$, then $\sum a_n$ **diverges**.

Warning: If $\lim_{n\to\infty} a_n = 0$, the test is **INCONCLUSIVE**. (The series may converge or diverge).

Special Series

Geometric Series: $\sum_{n=0}^{\infty} ar^n$
- **Converges** to $S = \frac{a}{1-r}$ if $|r| < 1$.
- **Diverges** if $|r| \geq 1$.

p-Series: $\sum_{n=1}^{\infty} \frac{1}{n^p}$
- **Converges** if $p > 1$.
- **Diverges** if $p \leq 1$.

Harmonic Series: $\sum_{n=1}^{\infty} \frac{1}{n}$
- Always **diverges** (p-series with $p = 1$).

Telescoping Series:
- Most intermediate terms cancel out.
- Find S_k (use partial fractions if needed), then evaluate $\lim_{k\to\infty} S_k$.

Tests for Positive Series ($a_n > 0$)

Integral Test:
Let $f(x)$ be continuous, positive, decreasing for $x \geq 1$, and $a_n = f(n)$.
- $\sum a_n$ converges if $\int_1^{\infty} f(x)\,dx$ converges.
- $\sum a_n$ diverges if $\int_1^{\infty} f(x)\,dx$ diverges.

Tip: Ensure $f'(x) < 0$ to verify it is decreasing.

Direct Comparison Test (DCT):
Choose a known series $\sum b_n$.
- If $a_n \leq b_n$ and $\sum b_n$ conv $\Rightarrow \sum a_n$ **conv**.
- If $a_n \geq b_n$ and $\sum b_n$ div $\Rightarrow \sum a_n$ **div**.

Limit Comparison Test (LCT):
Choose a known series $\sum b_n > 0$.
Let $L = \lim_{n\to\infty} \frac{a_n}{b_n}$.
- If $0 < L < \infty$, both converge or both diverge.
- If $L = 0$ and $\sum b_n$ conv $\Rightarrow \sum a_n$ conv.
- If $L = \infty$ and $\sum b_n$ div $\Rightarrow \sum a_n$ div.

Tip: Great for rational/algebraic functions. Pick b_n by taking leading terms of numerator and denominator.

Tests for General Series

Alternating Series Test (AST):
Series of the form $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ ($b_n > 0$).
Converges if BOTH conditions are met:
1. $b_{n+1} \leq b_n$ for all n (decreasing).
2. $\lim_{n\to\infty} b_n = 0$.

Ratio Test:
Let $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$.
- $L < 1 \Rightarrow$ **Absolutely Convergent**.
- $L > 1 \Rightarrow$ **Divergent**.
- $L = 1 \Rightarrow$ **Inconclusive**.

Tip: Indispensable for factorials ($n!$) and exponentials (r^n). **Fails** on p-series and rational functions.

Root Test:
Let $L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$.
- $L < 1 \Rightarrow$ **Absolutely Convergent**.
- $L > 1 \Rightarrow$ **Divergent**.
- $L = 1 \Rightarrow$ **Inconclusive**.

Tip: Use when the entire term is raised to the n-th power (e.g., $(f(n))^n$).

Absolute vs. Conditional

Absolute Convergence:
$\sum |a_n|$ converges. *(If a series converges absolutely, it implies $\sum a_n$ converges ordinarily.)*

Conditional Convergence:
$\sum a_n$ converges, but $\sum |a_n|$ diverges.
Example: The Alternating Harmonic Series converges conditionally.

Strategy: Which Test to Use?

1. Check $\lim a_n$. If $\neq 0 \Rightarrow$ **Divergence Test**.
2. Known series?
   - $ar^n \Rightarrow$ **Geometric Test**.
   - $1/n^p \Rightarrow$ **p-Series Test**.
3. Alternating signs? $\Rightarrow$ **AST**.
4. Has factorials/exponentials? $\Rightarrow$ **Ratio Test**.
5. Form is $(\dots)^n$? $\Rightarrow$ **Root Test**.
6. Looks like a p-series/geometric? $\Rightarrow$ **LCT / DCT**.
7. Function easily integrated? $\Rightarrow$ **Integral Test**.

Power Series

Form: $\sum_{n=0}^{\infty} c_n (x-a)^n$ centered at $x = a$.

Radius of Convergence (R): Distance from center where series converges. Find by solving:

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$$

using the Ratio Test on the absolute value.

Interval of Convergence (IoC): The domain $(a - R, a + R)$ where the series converges.
Crucial Step: You **MUST** test the endpoints $x = a - R$ and $x = a + R$ individually using standard convergence tests.

Common Maclaurin Series

$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots = \sum_{n=0}^{\infty} x^n$$

IoC: $(-1, 1)$

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

IoC: $(-\infty, \infty)$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

IoC: $(-\infty, \infty)$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

IoC: $(-\infty, \infty)$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}$$

IoC: $(-1, 1]$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$$

IoC: $[-1, 1]$

Error Bounds / Remainders

Alternating Series Estimation Theorem:
If $\sum (-1)^{n-1} b_n$ converges by AST, the error R_n when approximating with S_n is bounded by the first unused term:

$$|R_n| = |S - S_n| \leq b_{n+1}$$

Taylor's Inequality (Lagrange Error):
If $|f^{(n+1)}(t)| \leq M$ for all t between a and x:

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}$$